Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – PERIODIC TEST 12
Chapter 1: Real Numbers

Time: 60 Minutes
MM: 30

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based question of 4 marks with internal subparts.
7. Use of calculators is strictly prohibited. All mathematical steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. What is the greatest possible uniform walking speed at which a man can cover distances of 52 km and 91 km in an exact integer number of hours? [1]

(a) 7 km/h (b) 13 km/h (c) 17 km/h (d) 26 km/h

Q2. The prime factorisation of 3825 is: [1]

(a) $3 \times 5^2 \times 17$ (b) $3^2 \times 5^2 \times 17$
(c) $3^3 \times 5 \times 17$ (d) $3^2 \times 5^3 \times 17$

Q3. The prime factorisation of 5005 is: [1]

(a) $5 \times 7 \times 11 \times 13$ (b) $5^2 \times 7 \times 13$
(c) $5 \times 11^2 \times 13$ (d) $5 \times 7^2 \times 11$

Q4. The prime factorisation of 7429 is: [1]

(a) $13 \times 17 \times 23$ (b) $17 \times 19 \times 23$
(c) $17 \times 19 \times 29$ (d) $19 \times 23 \times 29$

Q5. If a and b are two consecutive natural numbers, then HCF(a, b) is: [1]

(a) 0 (b) 1 (c) 2 (d) ab

Q6. **Assertion (A):** $\frac{2\sqrt{45}+3\sqrt{20}}{2\sqrt{5}}$ is a rational number.
Reason (R): The square root of every positive integer is an irrational number. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. Find the HCF and LCM of 12, 15, and 21 by applying the prime factorisation method. [2]

Q8. Can two numbers have 15 as their HCF and 175 as their LCM? Give reasons. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. A rectangular courtyard 3.78 m long and 5.25 m broad is to be paved exactly with square tiles of the same size. Find the largest size of such tile and the minimum number of tiles required. [3]

Q10. The traffic lights at three different road crossings change after every 48 seconds, 72 seconds, and 108 seconds respectively. If they change simultaneously at 7:00:00 AM, at what time will they change simultaneously again? [3]

SECTION D: Long Answer Type (2 × 5 = 10 Marks)

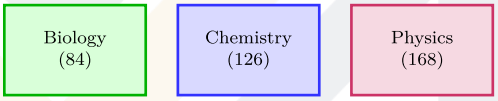
Q11. Find the smallest positive integer which when divided by 16, 20, and 24 leaves remainders 11, 15, and 19 respectively, and is also exactly divisible by 7. [5]

Q12. Let a and b be two positive integers. (i) If $\text{HCF}(a, b) = 18$ and $\text{LCM}(a, b) = 1080$, and one of the numbers is 216, find the other number. (ii) Can two numbers have 16 as their HCF and 380 as their LCM? Justify mathematically. (iii) If p and q are distinct prime numbers, find $\text{HCF}(p^3q^2, p^2q^4)$ and $\text{LCM}(p^3q^2, p^2q^4)$, and verify that their product equals the product of the two given expressions. [5]

SECTION E: Case Study Based Question (1 × 4 = 4 Marks)

Q13. CASE STUDY: CASE STUDY 12: SCIENCE EXHIBITION PODIUM PROJECT DISPLAY LOGISTICS [4 Marks]

A CBSE regional science exhibition is held in an auditorium. Student exhibits are registered in three branches: Biology (84 exhibits), Chemistry (126 exhibits), and Physics (168 exhibits). The organizing committee sets up exhibition aisles such that each row contains the exact same number of project stalls, every row contains exhibits from only one branch of science, and the number of stalls per row is maximized.



Max Stalls Per Row = HCF$(84, 126, 168) = 42$

(i) Find the maximum number of exhibition stalls in each row. [2]

(ii) Calculate the total number of rows required in the auditorium. [1]

(iii) Find the LCM of 84 and 126 by prime factorisation. [1]

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